

# Enyu Zhu

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## EDUCATION

**Massachusetts Institute of Technology - B.S. Artificial Intelligence, Architectural Design** Sep 2024 - May 2028  
Coursework: Machine Learning, Deep Learning, Probability & Statistics, Algorithms, UX Design GPA: 5.0/5.0

## EXPERIENCE

**Machine Learning Science Intern - Hostelworld, Porto, Portugal** Jun 2026 - Present

- Developed multimodal embeddings from 800K+ profiles, improving previous Meet Travelers pipeline by 80%
- Built recommendation models combining profile, behavioral, and visual signals to predict traveler compatibility
- Designed vector-search architecture replacing millions of pairwise comparisons with real-time recommendation
- Analyzed messaging, chat initiation, and engagement data across travelers in 180+ countries to optimize rankings

**Machine Learning Engineering Intern - Onsite Copilot, Cambridge, MA** Jan 2026 - Apr 2026

- Organized 1000+ construction documents and schedules into knowledge graphs improving retrieval accuracy by 30%
- Fine-tuned LLMs and VLMs on construction datasets to detect project rework and coordination errors
- Built multimodal retrieval systems combining site images and video data, generating embeddings for agent reasoning
- Integrated frontend interface for AI assistants, enabling image uploads and conversational interactions

**Data Analytics Intern - National University of Singapore Urban Analytics Lab, Singapore** Jun 2025 - Sep 2025

- Built PROSPER, a restaurant success prediction framework using LightGBM and contrastive FAISS retrieval
- Applied SHAP TreeExplainer across 47 pre-opening features, achieving 81% predictions within  $\pm 5$  score points
- Scraped 565,000+ Google Reviews, constructing a sentiment-based Bayesian-shrunk success label from review text
- Designed a RAG layer, reducing location intelligence cost from SGD 150–400/hour to SGD 0.01 per query

## RESEARCH

**SERC Scholar - MIT Schwarzman College of Computing, Cambridge, MA** Sep 2024 - Apr 2026

- Built a website, MetaCookies, using HTML/CSS to enhance public awareness on misinformation and deepfakes
- Designed non-consensual intimate imagery, AI image generation, and deepfake chatbot case studies
- Web-scraped environmental footprint, labor rights, and privacy policy data across 30+ companies in the S&P 500
- Presented at MIT Ethics of Computing Research Symposium and received funding for project continuation

**VR Researcher - CSAIL Human Computer Interaction, Cambridge, MA** Sep 2025 - Dec 2025

- Developed RepAR, a Unity-based VR system to repair fabrication models, reducing manual correction steps
- Integrated OpenXR and C++ frameworks to detect surfaces and create user-controlled pen in passthrough mode
- Evaluated VR workflows across 50+ fabrication models from Blender and Cura, identifying usability bottlenecks
- Validated repairs across FDM, SLA, and SLS fabrications to ensure compatibility for physical production constraints

**Simulation Researcher - MIT Media Lab City Science, Cambridge, MA** Feb 2025 - Jun 2025

- Integrated ReAct Agent in simulating different urban lifestyles in San Francisco for efficient zoning and urban plans
- Translated 30 resident interviews into narrative storylines in Figma to enhance behavioral realism in simulations
- Implemented interactive feedback stations in San Francisco to capture real-time input on urban interventions
- Customized interface features, aesthetics, transitions, and data collection with HTML/CSS

**Optimization Researcher - Carstensen Group, Cambridge, MA** Sep 2024 - Dec 2024

- Expanded HiTop 2.0's interactive features into an Unreal VR environment for physical interaction and live feedback
- Conducted user testing on VR prototypes to improve optimization workflow visualization and interface usability
- Developed C++ frameworks, Blueprints, and Flipbooks to control Quest3's handtracking in application
- Built an Unreal-to-MATLAB pipeline to support live transfer of joystick movements into optimization calculations

## SKILLS

- **Technical:** Python, C++, SQL, HTML/CSS, RAG, VR/AR, RL, R, Ranking, Recommendations, Web Scraping
- **Design:** Figma, Illustrator, Blender, Unity, Unreal Engine, 3D Printing
- **Languages:** Mandarin, Cantonese, Russian